Step 1: Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Load Dataset

```
data= pd.read_csv("BigBasket Products.csv")
```

```
data.head(12)   #Use head function to look for first 12 rows
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
0	1	Garlic Oil - Vegetarian Capsule 500 mg	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda	220.0	220.0	Hair Oil & Serum	4.1	This Product contains Garlic Oil that is known...
1	2	Water Bottle - Orange	Kitchen, Garden & Pets	Storage & Accessories	Mastercook	180.0	180.0	Water & Fridge Bottles	2.3	Each product is microwave safe (without lid), ...
2	3	Brass Angle Deep - Plain, No.2	Cleaning & Household	Pooja Needs	Trm	119.0	250.0	Lamp & Lamp Oil	3.4	A perfect gift for all occasions, be it your m...
3	4	Cereal Flip Lid Container/Storage Jar - Assort...	Cleaning & Household	Bins & Bathroom Ware	Nakoda	149.0	176.0	Laundry, Storage Baskets	3.7	Multipurpose container with an attractive desi...
4	5	Creme Soft Soap - For Hands & Body	Beauty & Hygiene	Bath & Hand Wash	Nivea	162.0	162.0	Bathing Bars & Soaps	4.4	Nivea Creme Soft Soap gives your skin the best...
5	6	Germ - Removal Multipurpose Wipes	Cleaning & Household	All Purpose Cleaners	Nature Protect	169.0	199.0	Disinfectant Spray & Cleaners	3.3	Stay protected from contamination with Multipu...
6	7	Multani Mati	Beauty & Hygiene	Skin Care	Satinance	58.0	58.0	Face Care	3.6	Satinance multani matti is an excellent skin t...
7	8	Hand Sanitizer - 70% Alcohol Base	Beauty & Hygiene	Bath & Hand Wash	Bionova	250.0	250.0	Hand Wash & Sanitizers	4.0	70%Alcohol based is gentle of hand leaves skin...
8	9	Biotin & Collagen Volumizing Hair Shampoo + Bi...	Beauty & Hygiene	Hair Care	StBotanica	1098.0	1098.0	Shampoo & Conditioner	3.5	An exclusive blend with Vitamin B7 Biotin, Hyd...
9	10	Scrub Pad - Anti-Bacterial, Regular	Cleaning & Household	Mops, Brushes & Scrubs	Scotch brite	20.0	20.0	Utensil Scrub-Pad, Glove	4.3	Scotch Brite Anti- Bacterial Scrub Pad thoroug...

Next steps: Generate code with data New interactive sheet

STEP 3: Description of the data in the DataFrame.

```
data.describe()
```

	index	sale_price	market_price	rating
count	27555.00000	27549.000000	27555.000000	18919.000000
mean	13778.00000	334.648391	382.056664	3.943295
std	7954.58767	1202.102113	581.730717	0.739217
min	1.00000	2.450000	3.000000	1.000000
25%	6889.50000	95.000000	100.000000	3.700000
50%	13778.00000	190.320000	220.000000	4.100000
75%	20666.50000	359.000000	425.000000	4.300000
max	27555.00000	12475.000000	12500.000000	5.000000

STEP 4: Information about the DataFrame.

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27555 entries, 0 to 27554
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   index         27555 non-null  int64  
 1   product       27554 non-null  object 
 2   category      27555 non-null  object 
 3   sub_category  27555 non-null  object 
 4   brand         27554 non-null  object 
 5   sale_price    27549 non-null  float64
 6   market_price  27555 non-null  float64
 7   type          27555 non-null  object 
 8   rating        18919 non-null  float64
 9   description   27440 non-null  object 
dtypes: float64(3), int64(1), object(6)
memory usage: 2.1+ MB
```

STEP 5: Top & Least Sold Products

```
data["product"].head()
```

	product
0	Garlic Oil - Vegetarian Capsule 500 mg
1	Water Bottle - Orange
2	Brass Angle Deep - Plain, No.2
3	Cereal Flip Lid Container/Storage Jar - Assort...
4	Creme Soft Soap - For Hands & Body

dtype: object

```
data["product"].tail()
```

	product
27550	Wottagirl! Perfume Spray - Heaven, Classic
27551	Rosemary
27552	Peri-Peri Sweet Potato Chips
27553	Green Tea - Pure Original
27554	United Dreams Go Far Deodorant

dtype: object

STEP 6: Measuring discount on a certain item.

```
data['discount'] = data['market_price'] - data['sale_price']
data['discount_percent'] = (data['discount'] / data['market_price']) * 100
data[['product', 'brand', 'market_price', 'sale_price', 'discount_percent']].head()
```

	product	brand	market_price	sale_price	discount_percent
0	Garlic Oil - Vegetarian Capsule 500 mg	Sri Sri Ayurveda	220.0	220.0	0.000000
1	Water Bottle - Orange	Mastercook	180.0	180.0	0.000000
2	Brass Angle Deep - Plain, No.2	Trm	250.0	119.0	52.400000
3	Cereal Flip Lid Container/Storage Jar - Assort...	Nakoda	176.0	149.0	15.340909
4	Creme Soft Soap - For Hands & Body	Nivea	162.0	162.0	0.000000

Step 7 : Missing Values from the Dataset.

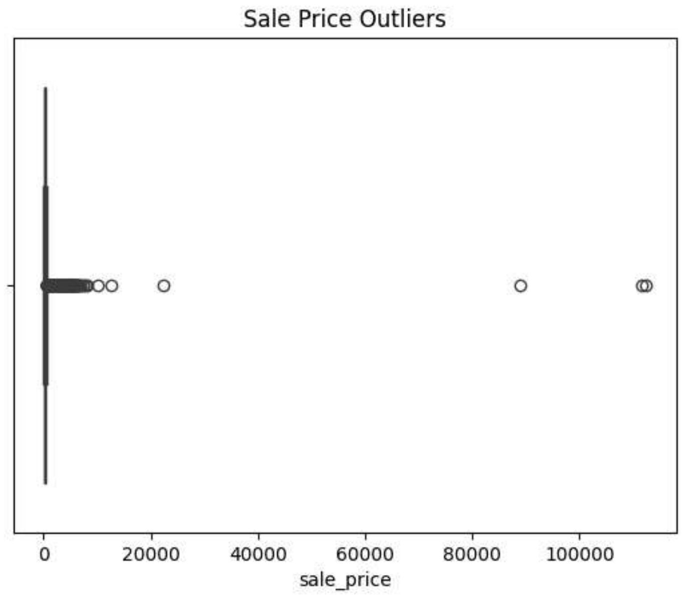
```
data.isnull().sum()
```

	0
index	0
product	1
category	0
sub_category	0
brand	1
sale_price	6
market_price	0
type	0
rating	8636
description	115
discount	6
discount_percent	6

dtype: int64

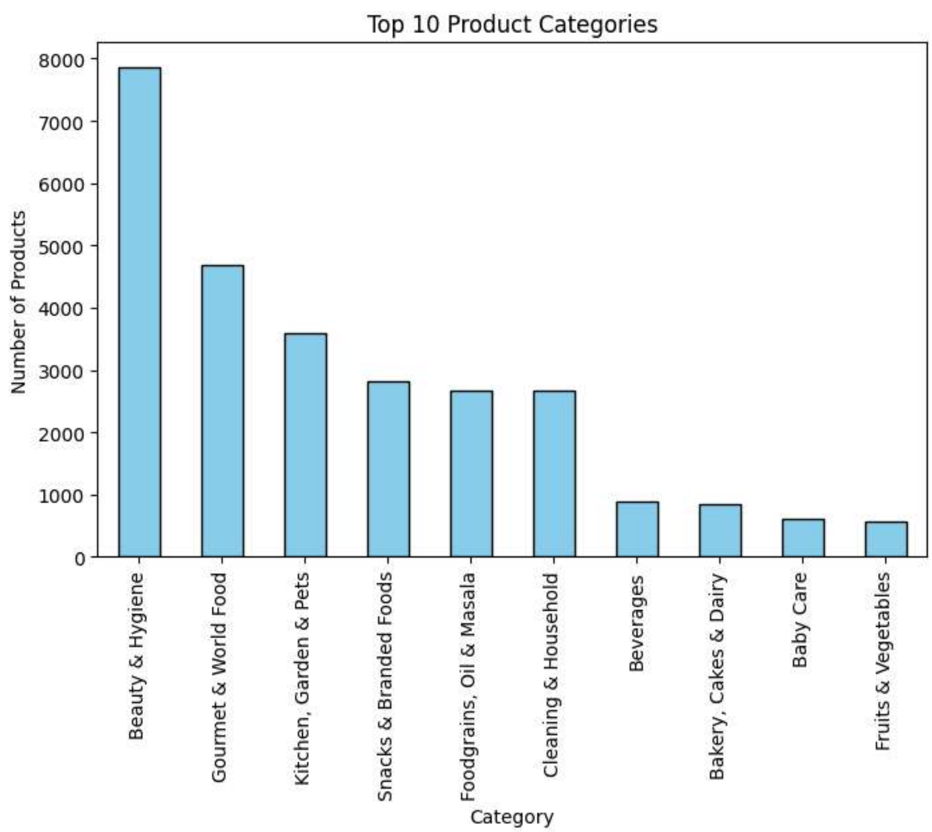
STEP 8:outliers from the dataset.

```
sns.boxplot(x=data['sale_price'])
plt.title("Sale Price Outliers")
plt.show()
```



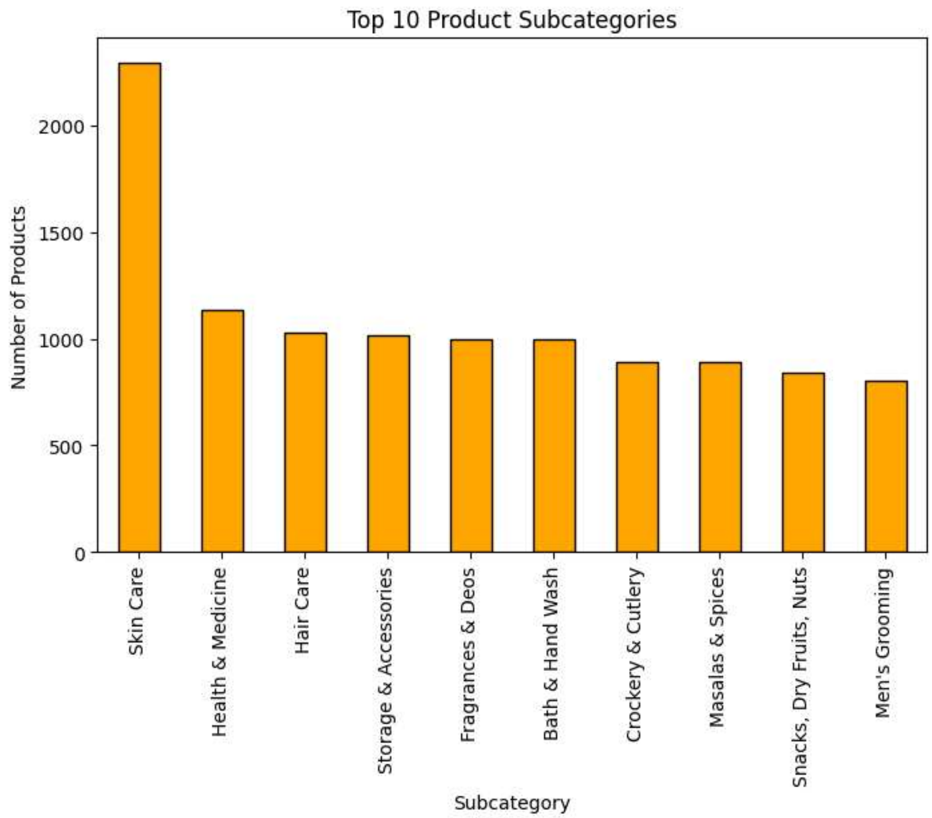
Step 9: Create Plots or visualizations.

```
plt.figure(figsize=(8,5))       #Category Distribution
data['category'].value_counts().head(10).plot(kind='bar', color='skyblue', edgecolor='black')
plt.title('Top 10 Product Categories')
plt.xlabel('Category')
plt.ylabel('Number of Products')
plt.show()
```




```
plt.figure(figsize=(8,5))      #Subcategory Distribution
data['sub_category'].value_counts().head(10).plot(kind='bar', color='orange', edgecolor='black')
plt.title('Top 10 Product Subcategories')
plt.xlabel('Subcategory')
```

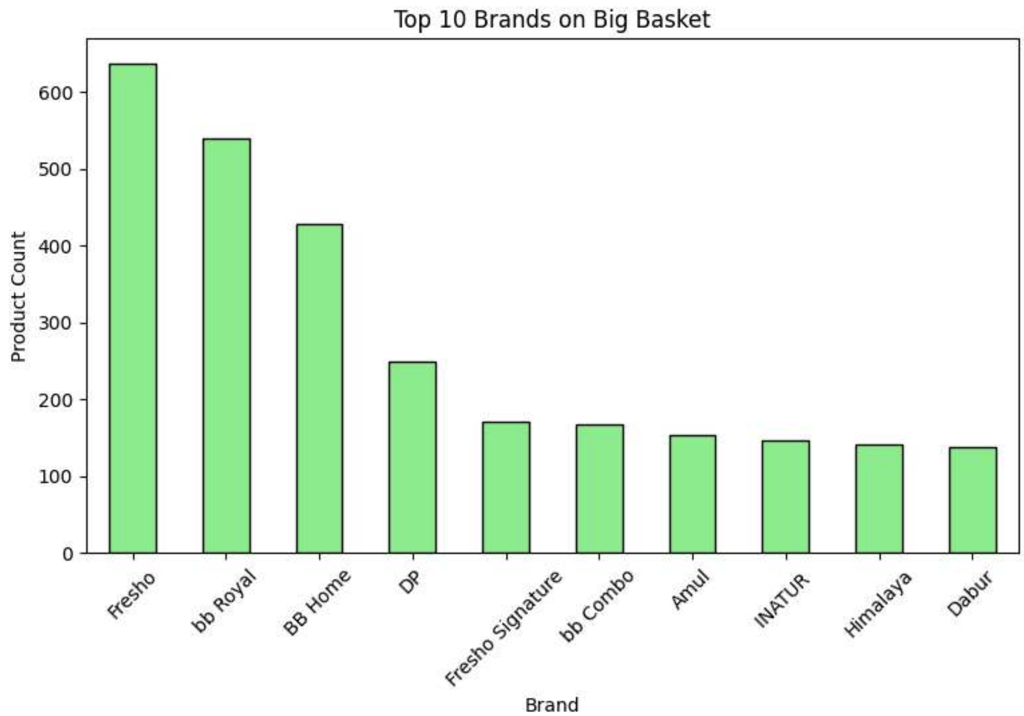
```
plt.ylabel('Number of Products')
plt.show()
```




```
plt.figure(figsize=(6,5))          #Price Comparison (Sale vs Market Price)
sns.scatterplot(x='market_price', y='sale_price', data=data, color='purple')
plt.title('Market Price vs Sale Price')
plt.xlabel('Market Price')
plt.ylabel('Sale Price')
plt.show()
```

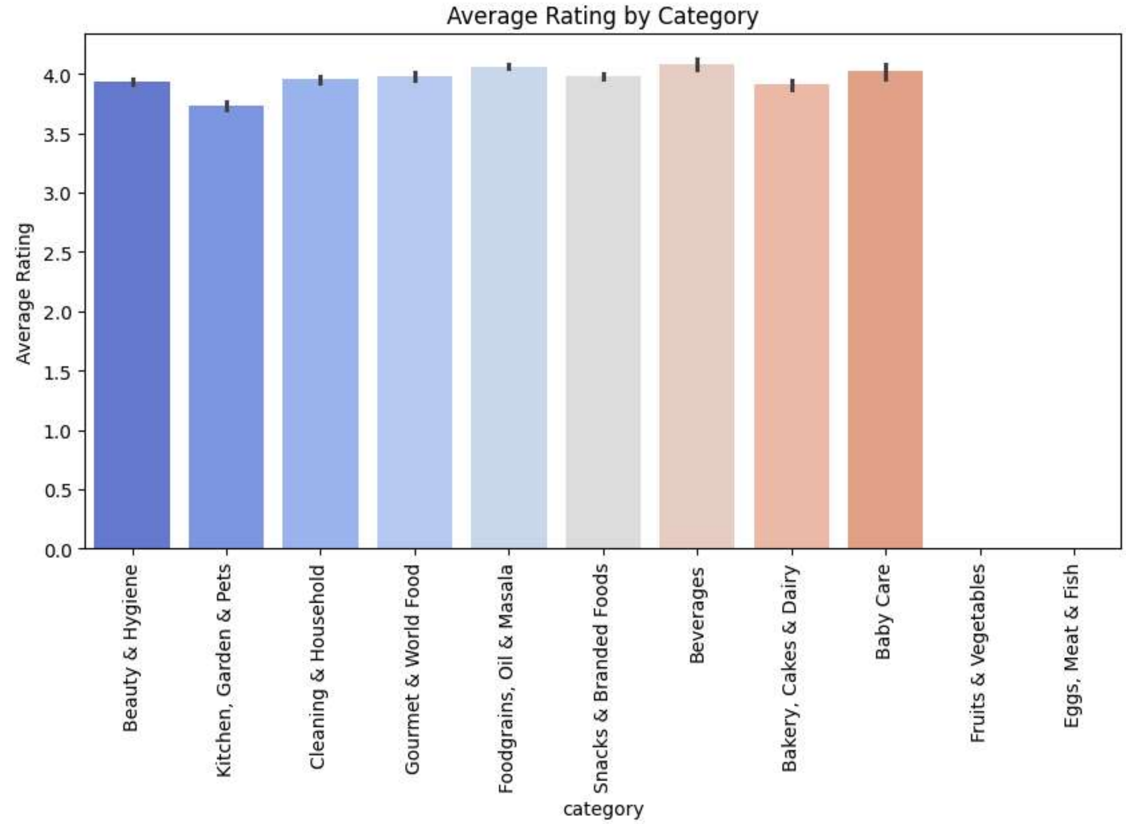

```
plt.figure(figsize=(9,5))
data['brand'].value_counts().head(10).plot(kind='bar', color='lightgreen', edgecolor='black')
plt.title('Top 10 Brands on Big Basket')
plt.xlabel('Brand')
```

```
plt.ylabel('Product Count')
plt.xticks(rotation=45)
plt.show()
```




```
plt.figure(figsize=(10,5))
sns.barplot(x='category', y='rating', data=data, palette='coolwarm')
plt.title('Average Rating by Category')
plt.xticks(rotation=90)
plt.ylabel('Average Rating')
plt.show()
```

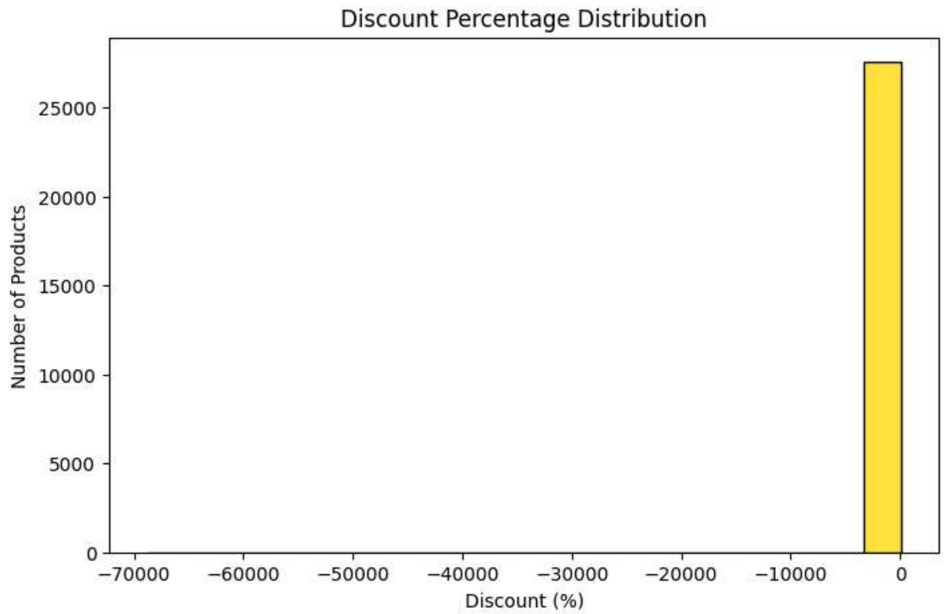
```
/tmp/ipython-input-1090525941.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `le

  sns.barplot(x='category', y='rating', data=data, palette='coolwarm')
```



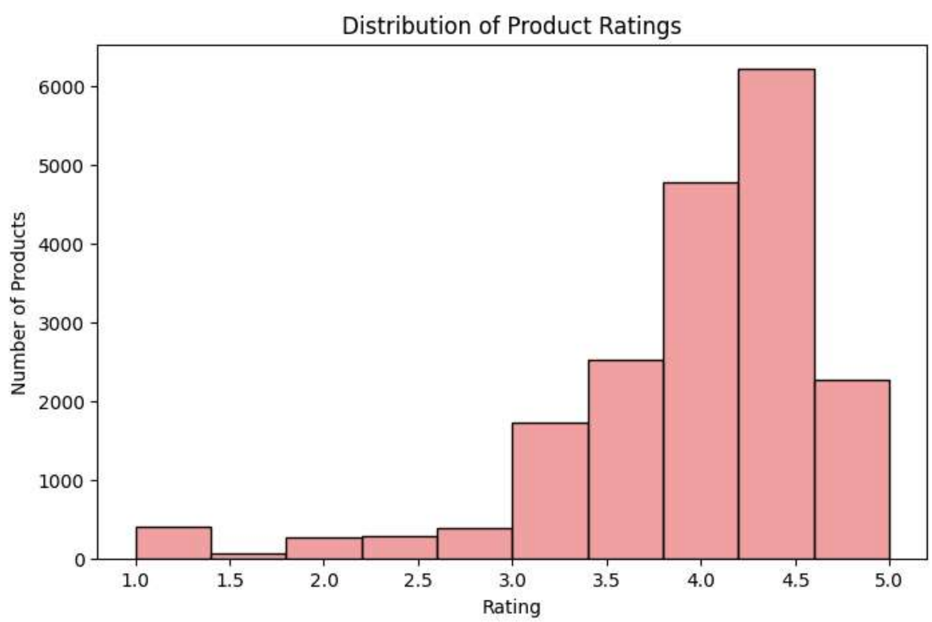

```
plt.figure(figsize=(7,5))
sns.scatterplot(x='market_price', y='sale_price', data=data, color='red', alpha=0.6)
plt.title('Market Price vs Sale Price')
plt.xlabel('Market Price')
plt.ylabel('Sale Price')
plt.show()
```


```
plt.figure(figsize=(8,5))
sns.histplot(data['discount_percent'], bins=20, color='gold', edgecolor='black')
```

```
plt.title('Discount Percentage Distribution')
plt.xlabel('Discount (%)')
plt.ylabel('Number of Products')
plt.show()
```




```
plt.figure(figsize=(8,5))
sns.histplot(data['rating'], bins=10, color='lightcoral', edgecolor='black')
plt.title('Distribution of Product Ratings')
plt.xlabel('Rating')
plt.ylabel('Number of Products')
plt.show()
```




```
plt.figure(figsize=(10,5))
sns.barplot(x='category', y='sale_price', data=data, palette='pastel')
plt.title('Average Sale Price by Category')
plt.ylabel('Average Sale Price')
plt.show()
```

```
/tmp/ipython-input-508119108.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `le

  sns.barplot(x='category', y='sale_price', data=data, palette='pastel')
```